

AUTONOMOUS ML RESEARCH / TECHJAM 2026

SciOdyssey

Long-Horizon Autonomous ML Research Agent

A persistent research world. A fresh Scientist every cycle.

PERSISTENT WORLD

FRESH SCIENTIST

EVIDENCE FIRST

Problem → System → Experience → Evaluation → Insights



Can an agent carry a research task to completion?

Turn a task contract into a tested model, a reproducible implementation, and evidence another researcher can inspect.

Explore

Choose hypotheses, write code, and run experiments.

Scientific judgment

Recover

Handle failures and revise unsuccessful ideas.

Long-running work

Deliver

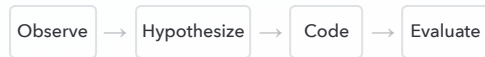
Retain a valid model and the evidence behind it.

Verifiable progress

Our test case: short-video ranking on KuaiRand-Pure, with a fixed evaluator and public-validation boundary.

Why research agents fall short

01 WORKFLOW

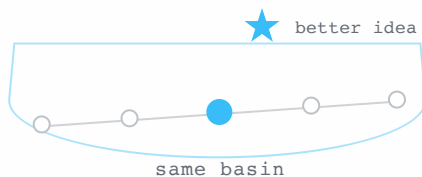


 **ERROR**
↓
dead end

Fixed recovery paths

Unexpected errors need human help.

02 SINGLE AGENT



History shapes search

Errors can persist

03 TREE SEARCH

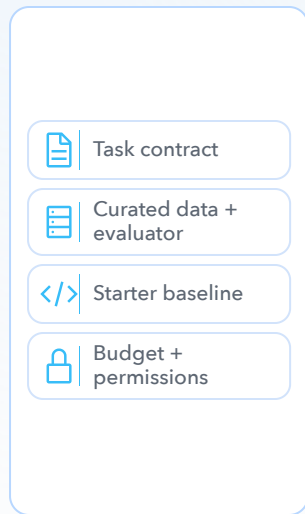


Repeated context reads

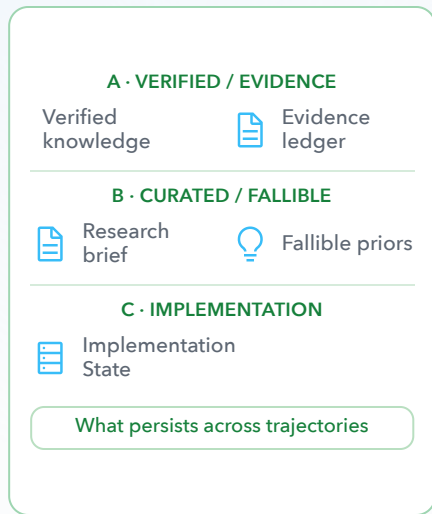
Each small edit pays the reading cost.

A system for sustained research

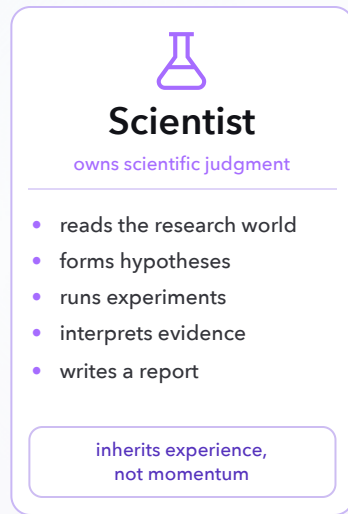
Environment



Research World



Fresh Scientist



META Review



Selective persistence / trajectory handoff

audit → scope → preserve

Coding-agent harness + runtime



Workspace isolation



Runner + evaluator



Logging



Artifact capture



State operations

Start with the task and working constraints.

The operator defines the objective and constraints; the agent chooses and executes the experiments.

01 task.md

What to solve.

Objectives · constraints · evaluation

EXAMPLE

Rank short videos for each user and interaction context.

KuaiRand-Pure · long_view · GAUC + nDCG@5

02 PERSONAL.md

How you want it solved.

Preferences · research style · priorities

EXAMPLE

Run on macOS / Apple Silicon with uv.

≤ 15 min / experiment · never print credentials

Autonomous by default. Supervisable by design.

Evidence changed the next experiment.

E001-E013: reject weak ideas, improve representations, then test ensembles.

E001	Screen historical target encodings; reject weak variants.	Medium screening
E003	Establish a valid rich-FM checkpoint.	Full · 0.6016310
E008	Retain a five-seed, 38-field FM ensemble.	Full · 0.6040901
E013	Select the eight-seed, 46-field FM ensemble.	Full · 0.6059363

These are milestones within four autonomous cycles. Medium screens ideas; Full evaluations support retained score claims.

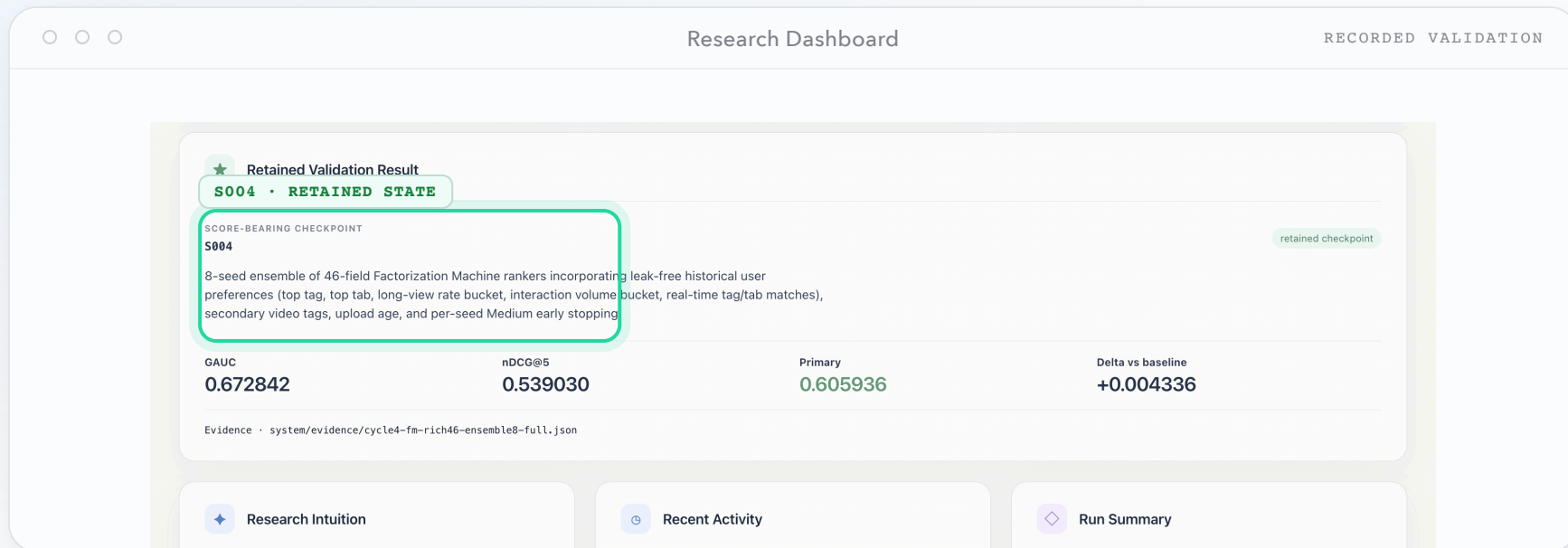
An evaluation failure became a recoverable step.

Cycle 1 completed training and evaluation, then failed while writing the evidence file.

Failure	NumPy float32 values broke JSON serialization.	Evidence writer
Repair	The Scientist converted scalars and reran.	Agent-authored fix
Retain	Printed measurements and saved evidence survived.	Auditable recovery

No human selected this repair or edited the code. This is one observed recovery case, not a guarantee for every failure.

A retained state keeps the evidence inspectable.



Inspect the retained state and the evidence behind it.

Retained validation checkpoint · evidence path · reproducible implementation.

Autonomous research. Measured gains.

PRIMARY / RESEARCH AGENT

0.6059363

+0.0043363 absolute improvement

METRIC	Official FM	Ours
GAUC	0.6674000	0.6728421
nDCG@5	0.5357000	0.5390304
Primary	0.6016000	0.6059363

Primary = mean(GAUC, nDCG@5). Public validation only; hidden-test scoring is organizer-controlled.

Final recipe: 46 categorical features × 8-seed FM ensemble · NumPy / CPU

An audit trail from experiment to output.

4

Autonomous cycles

Within a 50-iteration cap

13

Named experiments

E001-E013 · 7 Full evaluations

0

Manual scientific interventions

After launch · 0 GPU-hours

170,588 prediction rows passed the unchanged Starter Kit alignment checker.

LLM tokens: 48.24M including cache reads; 4.02M excluding cache reads.

Resource use is part of the evidence.

Project-level subscription estimate, with run-level telemetry reported separately.

~\$10

from first line of code to final result

coding · debugging · agent runs · all experiments

MODEL

One GPT Plus + One Gemini Pro for 7 days

Two consumer subscriptions covered the entire research loop.

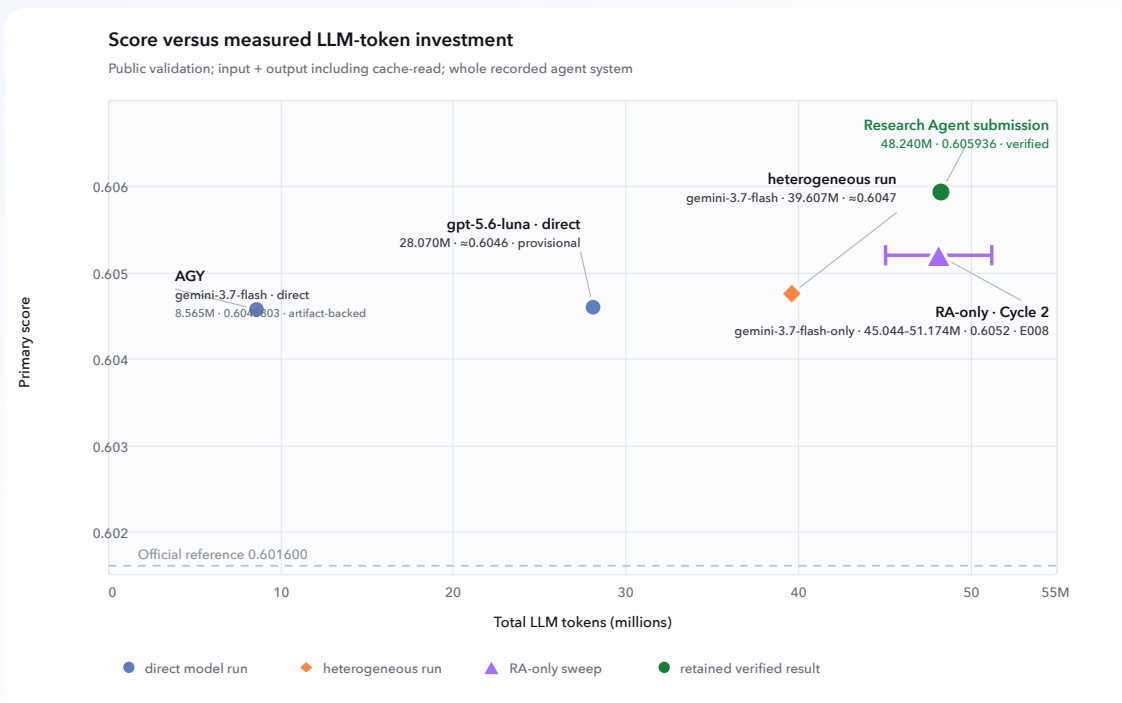
COMPUTE

1 × MacBook M2

All development and experiments ran on a single consumer laptop.

The recorded runs used CPU training; LLM usage remains a separate cost.

Compare scores alongside their evidence.



Runs differ in model, budget, and evidence quality; this is not a controlled causal comparison.

The run taught us more than the final score.

What survived the experiments – and what remains unsettled.

01 Diversity is a prior.

Keep `gemini-3.7-flash` as META while Scientist rotates across `gpt-5.6-sol`, `gemini-3.7-flash`, and `gpt-5.6-luna`. This widens priors; it does not prove the gain.

03 Failure can be evidence.

After evaluation, NumPy `float32` values broke serialization. The Scientist fixed the writer, reran, and kept the measurements.

02 A score is not a claim.

Audits found leakage and sampling mismatches in competing experiments. META can scope evidence; it is not a formal verifier.

04 Reset $\neq$ no anchor.

Fresh cognition reopens the search, but shared summaries can still omit context or anchor future work. Share evidence later than cognition.

OBSERVED, NOT PROVEN

Parallel/Synthesis reached public-validation Primary 0.6055536; the canonical E001-E013 frontier remains 0.6059363.

4 cycles

0 post-launch interventions

0 GPU-hours

Reproducible evidence. Open questions.

-
- | | | |
|----|---|-------------------|
| 01 | One task does not establish generality. | External validity |
|----|---|-------------------|
-
- | | | |
|----|--|--------------------|
| 02 | META, State, and reset are not isolated. | Causal attribution |
|----|--|--------------------|
-
- | | | |
|----|--|----------------|
| 03 | Summaries can lose context and anchor. | Memory quality |
|----|--|----------------|
-
- Next questions: broader tasks, targeted ablations, and less repeated execution.

A clean repo. A continuous agent.

FIELD NOTE / 01

Engineering practices to apply next: isolate parallel changes and make unattended work reviewable.

01 MANY AGENTS



Give every agent a branch.

main

clean trunk

● agent A / branch

● agent B / branch

● agent C / branch

review

→ merge

Let agents work independently. Conflicts stay local, and the main branch stays clean.

02 NO QUOTA / NO LAPTOP



Leave the next move in GitHub.

01

**Issue /
prompt**

inject the task



02

**GitHub
Action**

run the agent



03

Commitreview the
change

Connect the agent to GitHub Actions. It can code and commit while you are offline; review before merging.

Keep the experience. Reopen the search.

```
# From the configured repo, run one autonomous cycle
$ ./scripts/research-agent step --cli codex \
  --target /absolute/path/to/project --allow-edits
```

github.com/zc6600/research-agent-kuairand ↗

Code · Technical report · Experiment ledger · Checked output

Research is an **open-world** task.
We need an **open-world** coding agent.

01 OPEN ACTION SPACE



Shell



Files



Browser



Git

Use what exists

02 SELF-RECOVERY



Read errors



Inspect docs



Inspect
source



Install
packages

Fix what breaks

03 RUNTIME EXTENSION



MCP



Skills



CLI



Packages

Add what's missing

WORKFLOW EXAMPLE • LANGGRAPH

LangGraph makes a known workflow explicit as nodes + edges.

Missing capability? Leave the graph → inspect → extend → retry

Five people. One autonomous research loop.

OVERLAPPING RESPONSIBILITIES / SHARED WORLD



Chen Zhu

TEAM LEAD / SYSTEM

Direction · architecture ·
integration



Zhou Ziyu

MODELING / EXPERIMENTS

Features · ensembles ·
evidence review



Shilin Xu

DATA / METRICS

Data workflow · contracts
· alignment



GE GAO

RUNTIME / RELIABILITY

Execution · testing ·
integration support



Jiran Li

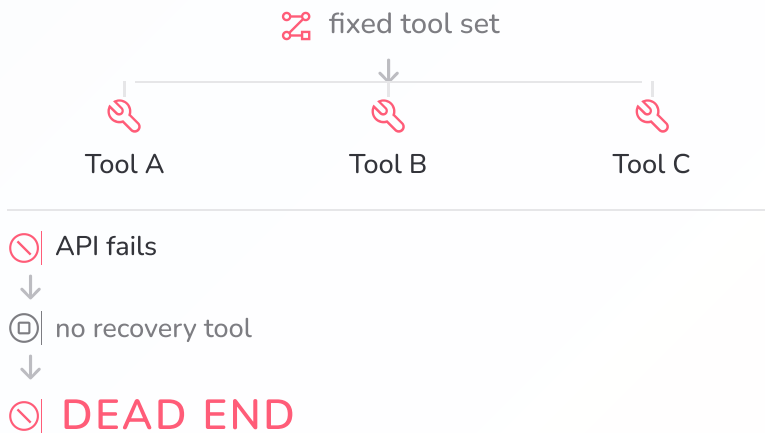
EVIDENCE /
REPRODUCIBILITY

Telemetry · documentation
· packaging

Research is an **open-world** task.
We need an **open-world** coding agent.

LANGGRAPH-STYLE ORCHESTRATION

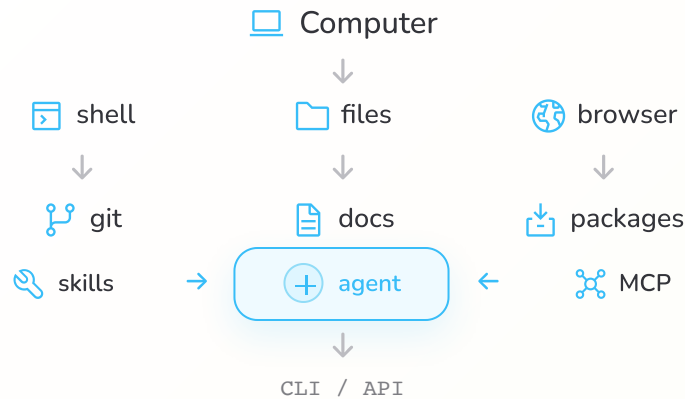
predefined graph



Capabilities are bounded by what developers anticipated.

CODING-AGENT HARNESS

open action space



fail → inspect → learn → modify environment → retry

The agent can acquire the missing capability at runtime.